

CEAMLS · HUMAN-AI COLLABORATION CHALLENGE

ReadAloud

A child reads for one minute.

The laptop scores it. Nothing leaves the room.

THE MINUTE THAT COSTS THE MOST

A teacher sits with one child while twenty seven others wait.

A running record is the most informative minute in elementary literacy assessment. Sixty seconds of reading, then four minutes of hand tallying every error. Three times a year, for every child.

The tallying is the part a machine should do. The judgement is not.

WHAT IT DOES

We already know what the passage says.

So this is not open transcription. It is alignment against a known text, which collapses the search space from forty thousand candidate words to about a hundred.

Microphone

›

Whisper, on device

›

Needleman–Wunsch alignment

›

Miscue classification

›

Running record

No server · No API key · No language model

NOW

Turn the wifi off and read to it.

Verified in a real browser with every external host blocked.

Record, transcribe, score, print. Zero outbound requests.

WHAT THE TEACHER SEES

Every deviation is named, and so is every forgiveness.

Nora ~~lost~~^o her red? mitten on the way to school^{sc}, but the bell^s rang for recess.

Green, dotted

Self correction. Not an error. Evidence the reader is monitoring themselves.

Red

Counted as an error. Struck through if the word was skipped, underlined if it was misread.

Amber

We are not confident enough to call it an error, so we do not.

THE CONTROL THAT MATTERS

The override runs in both directions.

Four filters sit in this engine and every one of them can only *remove* errors. If the only control a teacher had were the power to forgive, every adjustment the tool offered would push the score upward.

So a teacher can mark a flagged word correct, **and mark a forgiven word as an error.**

The teacher is the assessor. We are the stopwatch.

OUR HEADLINE FEATURE

A dialect layer, so a child who speaks African American English is not scored down for pronunciation.

Miscue analysis has done exactly that for decades. Speech recognisers make it worse: roughly double the word error rate for Black speakers across all five major commercial systems.

Labov & Baker 2010 · Charity et al. 2004 · Koenecke et al., PNAS 2020

WE MEASURED WHETHER IT FIRES

0

times, across five passages and four hundred and twenty one words.

We published the zero.

WHY

Whisper rewrites the dialect before our rules ever see it.

SPOKEN	WHISPER WROTE	
aksed	asked	normalised away
wif	with	normalised away
sista	sister	normalised away
dis col	Diskal	word boundary destroyed
fo fo	Fofo	word boundary destroyed

A layer that compares words cannot repair a broken word boundary. The dialect speaking reader is charged five errors on a twelve word passage, and our feature is blind to all of it.

SO THE HONEST CLAIM IS SMALLER, AND IT SURVIVES

The equity contribution is the architecture, not the rule table.

Aligning against a known hundred word passage collapses the recognition search space from roughly forty thousand candidates to a hundred. That absorbs most accent driven recognition error *structurally*, rather than by rule.

And no child's voice ever leaves the laptop.

WE TESTED IT

0.48%

FALSE POSITIVES, CLEAN READS

0.4

MEAN WCPM ERROR

77

AUTOMATED TESTS, ALL GREEN

MISCUE TYPE	PRECISION	RECALL	F1
Substitution	81.8%	90.0%	85.7%
Omission	93.8%	100.0%	96.8%
Insertion	83.3%	100.0%	90.9%
Self correction	100.0%	40.0%	57.1%

Ground truth is **constructed, not annotated**. We generate the spoken text from the reference by applying an explicit error script, so there is no labelling step to disagree with and no way to score our own homework.

WHY FALSE POSITIVES IS THE NUMBER WE OPTIMISED

Telling a child who read correctly that they made a mistake is the worse failure.

Those two errors do not trade off evenly in a classroom. One costs a teacher thirty seconds of checking. The other costs a seven year old some part of their belief that they can read.

WHAT IT CANNOT DO

We have never run this on a child's voice.

Every evaluation clip is speech synthesis, one adult voice reading faster than the ninetieth percentile for grade six. A seven year old at sixty words a minute with real pauses will behave differently, and that is the largest unknown in the system.

Self correction recall is forty percent by construction. Published norms stop at grade six. It is a screening aid and a research prototype, not a validated instrument.

The path to one is an inter rater reliability study against certified human scorers, stratified by dialect. That is a semester, not a weekend.

READALLOUD

No subscription.
No API key.
No child's voice leaving
the building.

It runs on hardware schools already own.

[Here is the printed running record.](#)